FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST– V
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	D
2.	C	A	A
3.	C	A	A
4.	A	A	D
5.	B	A	C
6.	D	A	A
7.	D	D	A
8.	C	C	D
9.	B, D	A, B, C	A, C
10.	A, B, D	A, B, C	B, C
11.	A, C	A, B, C	A, D
12.	A, B	C, D	A, B
13.	B	A	C
14.	D	D	D
15.	D	D	B
16.	A	A	C
17.	C	C	B
18.	B	B	D
1.	3	7	6
2.	9	4	9
3.	2	4	3
4.	5	4	1
5.	7	2	2

Physics

PART – I

SECTION – A

1. Apply conservation of energy and conservation of angular momentum between points at infinite separation and on the surface of the sphere.

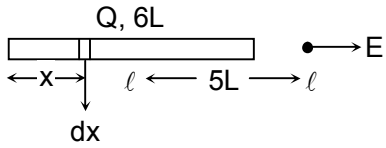
2. To prevent reflection $\frac{\lambda}{2} = 2\mu t$

$t = \frac{\lambda}{4\mu} = \frac{5.3\times10^{-7}}{4\times1.31} \simeq 0.1\mu m.$

3. All the corners of the hexagonal faces will be at same potential. So wires joining the corners can be removed. Then in the remaining circuit we will have six '3R' resistors connected in parallel between A and B.

4. E due to rod.

$= \int_0^{6L} \frac{1}{4\pi\in_0}\cdot\left(\frac{Q}{6L}\right)\frac{dx}{(8L-x)^2} = \frac{1}{64\pi\in_0}\frac{Q}{L^2}$



E_{net} = 2 E cos 30°.

5. $px^2 + qv^2 = r$

$\Rightarrow \quad 2px\frac{dx}{dt} + 2qv\frac{dv}{dt} = 0 \qquad \Rightarrow a = -\frac{p}{q}x$

$\Rightarrow$ Particle is executing S.H.M. with angular frequency $= \sqrt{\frac{p}{q}}$.

6. Particle will move in helical path having circular projection in y-z plane. In the given time it will complete half revolution.
So, $x = v_0t$, $y = 0$, $z = -2R$

7. $P = \left(\frac{E}{R+r}\right)^2.R$

P will be maximum when r = 0

8. $\frac{dV}{dt} = \frac{\pi D^2}{4}.\frac{dh}{dt}$

$\frac{dh}{dt} = \frac{4\times16.5}{3.14\times(6.30)^2} = 0.529 \text{ cm/s}$

9. $-1.5 = 15T - \frac{1}{2}(10+a)T^2$. Where T = 2 sec $\Rightarrow$ a = 5.75 m/s^2

As we can not find velocity of bullet with respect to ground, we can not calculate distance from the given data.

10. $Q_{BC} = nC_P\Delta T > 0$

$Q_{DA} = nC_V\Delta T < 0$

$\eta = \frac{W}{Q_{absorbed}} = 1 - \frac{Q_{released}}{Q_{absorbed}} = 1 - \frac{nC_V(T_D - T_A)}{nC_P(T_C - T_B)} = 1 - \frac{1}{\gamma}\left(\frac{T_D - T_A}{T_C - T_B}\right)$

11. Dipole moment will be angle of 30° to '–x' – axis of magnitude $= (\lambda.\pi a).\left(2\times\frac{2a}{\pi}\right) = 4\lambda a^2$.

E will be $2.\frac{2\lambda}{4\pi\varepsilon_0 a}$ at angle of 30° with '+x' – axis.

12. For $E_P = 0$

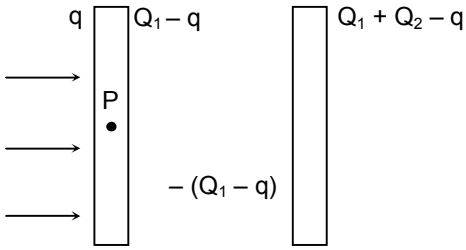
$$E + \frac{q}{2A\in_0} = \frac{Q_1 - q}{2A\in_0} + \frac{Q_2}{2A\in_0}$$

$$\Rightarrow \quad q = \frac{Q_1 + Q_2}{2} - \in_0 AE$$

E inside the gap = $E + \frac{Q_1 - Q_2}{2A\in_0}$

E on the left side of plates = $E - \frac{Q_1 + Q_2}{2A\in_0}$

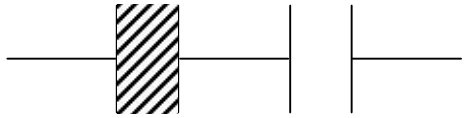
E on the right side of plates = $E + \frac{Q_1 + Q_2}{2A\in_0}$



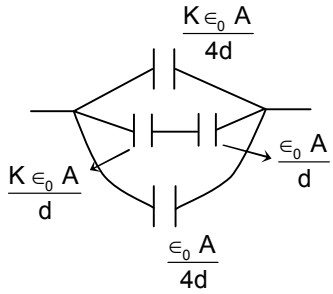
13. Apply conservation of Angular momentum about COM, to calculate final speed of the astronauts.
Work = Δ K.E. = $3\,Mv^2$

14. Torque about COM is zero hence angular momentum remains conserved

15. $$D = \frac{\left(\frac{K\in_0 A}{d/2}\right)\times\frac{\in_0 A}{d/2}}{\frac{k\in_0 A}{d/2} + \frac{\in_0 A}{d/2}}$$

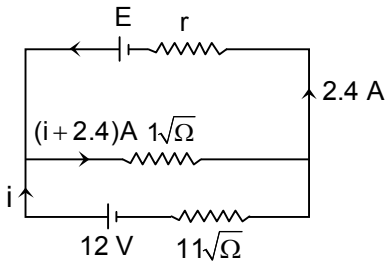


16.

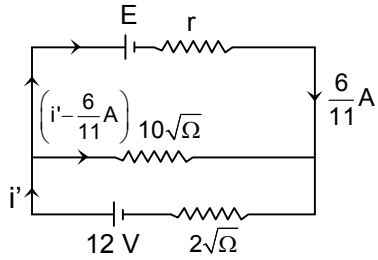


17-18.

First case :
12 = 12i + 2.4
$\Rightarrow$ i = 0.8 A
E = i + 2.4 (r + 1)
$\Rightarrow$ E = 3.2 + 2.4 r



Second case :



$12 = 12i^1 - \frac{60}{11}$

$\Rightarrow \quad i^1 = \frac{16}{11}A$

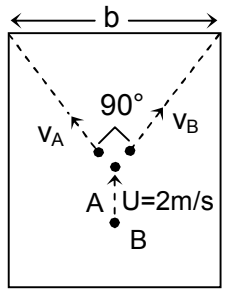
$E = 10i' - \frac{6}{11}(10 + r)$

$\Rightarrow \quad E = \frac{100}{11} - \frac{6r}{11}$ ……..(2)

Solving (1) and (2) $E = 8V, r = 2\Omega$

SECTION – C

1. $v_A^2 t^2 + v_B^2 t^2 = b^2$



when identical balls collide obliquely and collision is elastic and one of them at rest, they move mutually perpendicular to each other after collision.

From energy conservation

$\frac{1}{2}mu^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2$

so $b^2 = u^2 t^2 = 1$

$b = 1$ m

$\ell = 3$m

Area $= b\ell = 3m^2$.

2. $E_{n\to m} : E_{m\to 2} : E_{2\to 1} = 5 : 72 : 243$

$E_{m\to 2} \Rightarrow$ Transition corresponding to visible region.

$E_{n\to 2} : E_{2\to 1} = 77 : 243 \Rightarrow \left(\frac{1}{2^2} - \frac{1}{n^2}\right) : \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 77 : 243 \Rightarrow n = 9.$

3. For lens B according to the figure

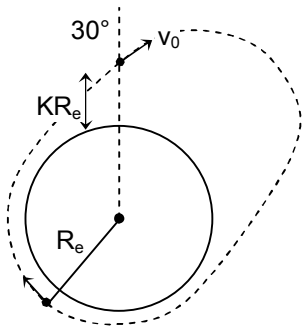
$u = +3$

$v = -6$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow f = -2$ m.

4. Conservation of angular momentum

$mv_0 \sin 30°(k+1)R_e = m\frac{\sqrt{15}}{4}v_e R_e$

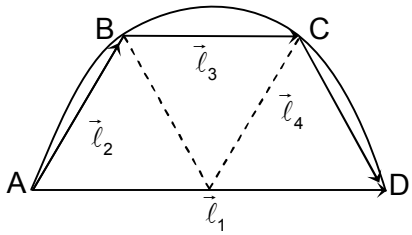


conservation of M.E.

$-\frac{GM_e m}{(k+1)R_e} + \frac{1}{2}mv_0^2 = -\frac{GM_e m}{R_e} + \frac{1}{2}m\left(\frac{15}{16}v_e^2\right)$

solving we get k = 5 or 9.

5. Given :



$F_{loop} = \left|i\overrightarrow{\ell_1} \times \vec{B}\right| = 8N$

$F_{AB} = \left|i\overrightarrow{\ell_2} \times \vec{B}\right| = 9N$

$\vec{F}_{BC} = \frac{1}{2}\vec{F}_{loop}$

Force on CD

$\vec{F}_{CD} = i\vec{\ell}_4 \times \vec{B} = i\left(\vec{\ell}_1 - \vec{\ell}_2 - \vec{\ell}_3\right) \times \vec{B}$

$= i\vec{\ell}_1 \times \vec{B} - i\vec{\ell}_2 \times \vec{B} - i\vec{\ell}_3 \times \vec{B}$

$= \vec{F}_{loop} - \vec{F}_{AB} - \vec{F}_{\frac{loop}{2}} = \vec{F}_{\frac{loop}{2}} - \vec{F}_{AB}$

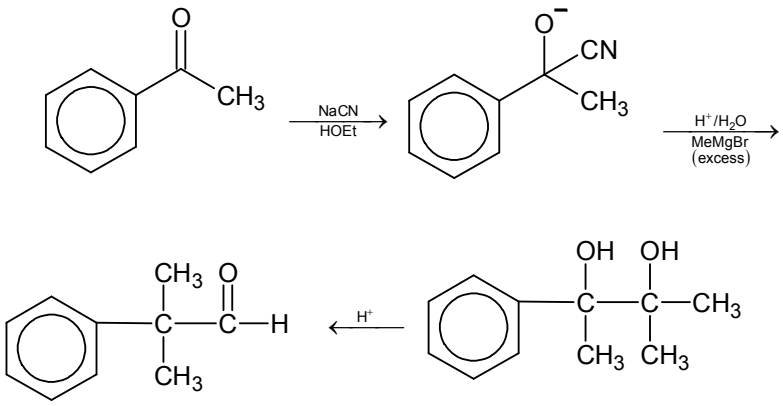
$F_{CD} = \sqrt{4^2 + 9^2 - 2 \times 4 \times 9 \times \frac{2}{3}} = 7$ N.

Chemistry

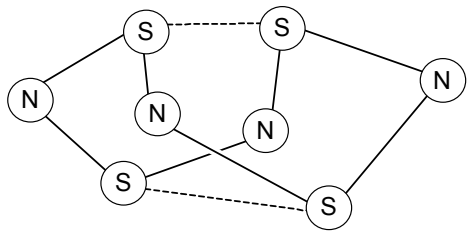
PART – II

SECTION – A

1. On removing atoms along the plane cutting two opposite edges.
In new NaCl lattice;
No. of Na^+ ion : No. of Cl^- atoms
$3Na^+$ atom : $\frac{5}{2}Cl^-$ atom.
In new fluorite structure:
No. of Ca^{2+} ion : No. of F^- ion
$\frac{5}{2}Ca^{2+} : 4F^-$
New cationic ratio $\frac{3}{\frac{5}{2}} = 6:5$
New anionic ratio = 5 : 8

3. 

4. $3Cu + 8HNO_3 \longrightarrow 2NO + 3Cu(NO_3)_2 + 4H_2O$
NO $\Rightarrow$ has 11 valence e^- with an unpaired.
e^- in π^*2p orbital.
$\Rightarrow$ is diamagnetic in solid state due to dimerisation.
$\Rightarrow$ shows resonance.

5. S_4N_4 

$\Rightarrow$4 five membered ring

6. If PF_3 is better π accepter than CO, than C – O bond order in II complex is more than I complex.

7. HNO_2 pale blue in colour due to dissolution of N_2O_3.
Reaction with:
(A) Br_2 water : $HNO_2 + H_2O + Br_2 \longrightarrow HNO_3 + 2HBr$
(B) $KMnO_4$: $2KMnO_4 + 3H_2SO_4 + 5HNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5HNO_3$

8. At stage I:

$$\underset{\text{Left}}{\underset{5\text{ meq}}{CH_3COOH} + \underset{0}{NaOH} \longrightarrow \underset{5\text{ meq}}{CH_3COONa}}$$

So, its is a buffer solution, [Salt] = [Acid]

$pH = pK_a \Rightarrow pH_I = 5$

At statge II:

$$\underset{\text{Left}}{\underset{0}{CH_3COOH} + \underset{0}{NaOH} \longrightarrow \underset{10\text{ meq in 200 ml.}}{CH_3COONa}}$$

So, c = 0.05

So, anionic hydrolysis takes place and pH is:

$$\& \ pH = 7 + \frac{1}{2}\left(pK_a + \log C\right)$$

$$pH_{II} = 7 + \frac{1}{2}\left(5 + \log 0.05\right) = 8.84$$

Stage III:

$$\underset{\text{Left}}{\underset{0}{CH_3COOH} + \underset{2\text{ meq/220ml}}{NaOH} \longrightarrow \underset{10\text{ meq/220 ml}}{CH_3COONa}}$$

So, pH will be according to NaOH.

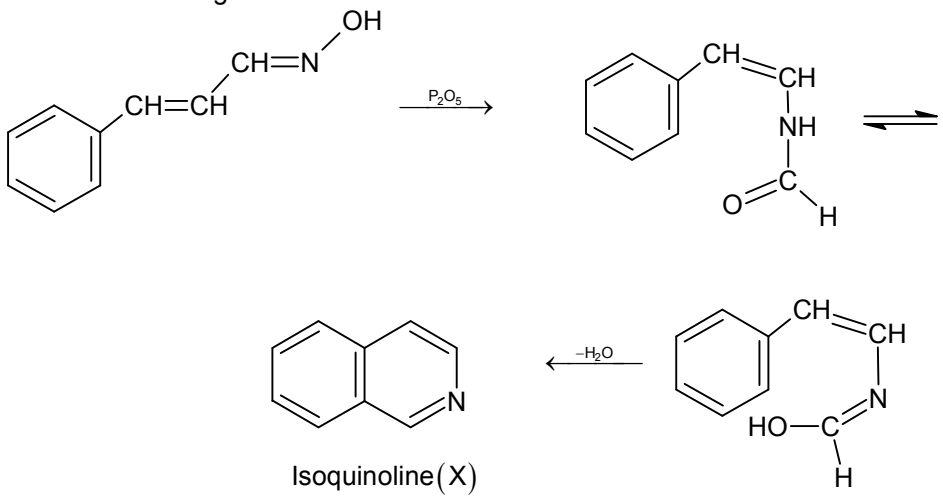
$[NaOH] = [a \times 10^{-3}]$

So, $[OH^-] = a \times 10^{-3}$

pOH = 2.040

$pH_{III} = 11.95$.

9. Beckmann rearrangement:



Isoquinoline(X)

(B) is aromatic, more basic than aniline.

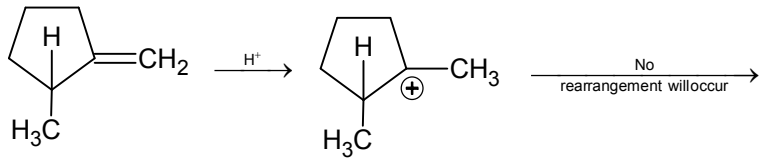
10. $CH_3NH_2 < (CH_3)_2NH_2^+$

$PCl_2F_3 < \underset{(\theta=180^\circ)}{PCl_3F_2}$

$\underset{(B.O=3)}{NO^+} > \underset{(B.O.=2.5)}{NO} \Rightarrow NO^+ < NO$

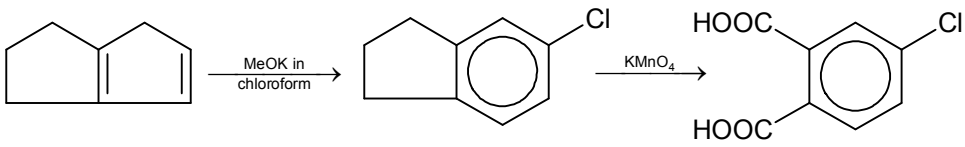
$\overset{+6}{Cr}O_3 > \overset{+3}{Cr}_2O_3$ (Acidic nature)

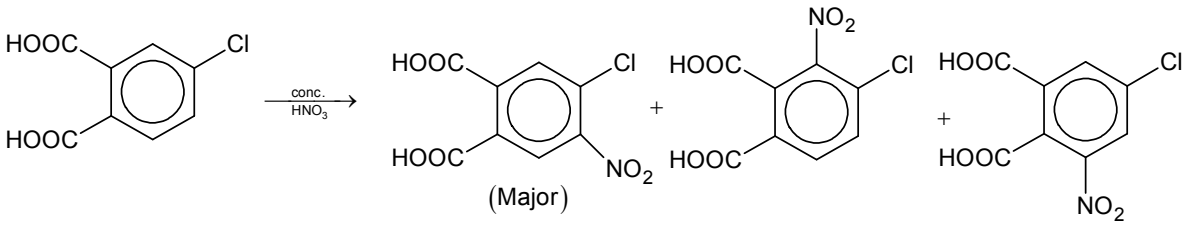
11. (A) $CD_3-CH(OH)-CH_3 \xrightarrow{conc.H_2SO_4} \underset{(Major)}{CD_3-CH=CH_2} + \underset{(Minor)}{CD_2=CH-CH_3}$

(C) 

(D) 1, 3 ⇒ diequitorial ⇒ *cis*
1,3 ⇒ axial and equatorial = 1 *trans*
So, groups are more stable at equatorial position.

12. Electrode reactions are:
Anodic reaction⇒
$Pb(s)+HSO_4^-(aq.) \longrightarrow PbSO_4(s)+H^+(aq.)+2e^-$
Cathodic reaction⇒
$PbO_2(s)+HSO_4^-(aq.)+3H^+(aq.)+2e^- \longrightarrow PbSO_4(s)+2H_2O(\ell)$
Net reaction: $Pb(s)+PbO_2(s)+2H_2SO_4(aq.) \longrightarrow PbSO_4(s)+2H_2O(\ell)$

15. 

16. 

Solution for the Q. No. 17 to 18.

$\underset{(A)}{ZnS}+H_2SO_4(dil.) \longrightarrow \underset{(C)}{ZnSO_4}+\underset{(B)}{H_2S}\uparrow$

Eq. (2) $3H_2S+K_2Cr_2O_7+4H_2SO_4 \longrightarrow K_2SO_4+Cr_2(SO_4)_3+7H_2O+\underset{(D)}{3S}$

$\Rightarrow (D)+O_2 \longrightarrow \underset{(E)}{SO_2}\uparrow \xrightarrow{2H_2S(B)} \underset{(colourless\ liq.)}{2H_2O} + \underset{(D)}{3S}\downarrow$

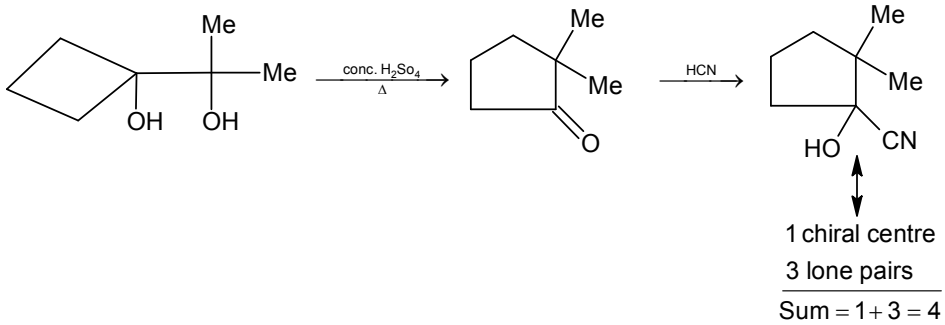
$\downarrow$

$\underset{(Blue)}{CuSO_4.5H_2O}$

$\underset{(C)}{ZnSO_4}+2NaOH \longrightarrow Zn(OH)_2\downarrow \xrightarrow{2NaOH} \underset{(Soluble)}{Na_2ZnO_2}+2H_2O$

In Eq. (2) $n_{eq}(H_2S) = n_{eq}(K_2Cr_2O_7)$
Moles(H_2S) × 2 = 0.2 × 10 × 6 = 6 milli moles
⇒ wt. of ZnS required is ⇒ Co = (6 × 97.3) mg
= 583.8 mg.

SECTION – C

2. 

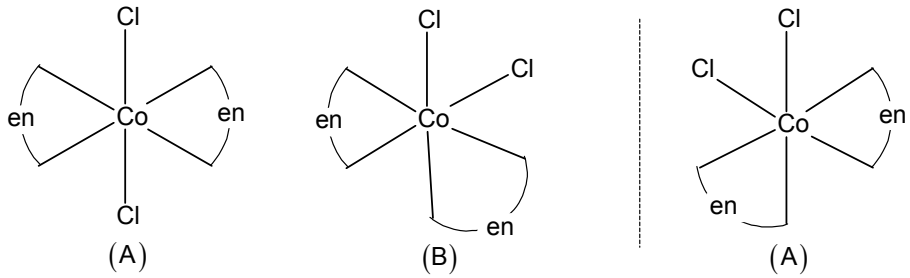
1 chiral centre
3 lone pairs
Sum = 1 + 3 = 4

3. $q = \Delta U = w = 0 - w = -w$
[-2(16 – 4) + (-1 × (32 – 16)] = 4 kJ.

4. Solid $PCl_5 \rightarrow [PCl_4]^+ [PCl_6]^-$
(sp^3, sp^3d^2)

Solid $PBr_5 \rightarrow [PBr_4]^+ [Br]^-$
(sp^3)

Solid $N_2O_5 \rightarrow [NO_2^+]^+ [NO_3^-]^-$
(sp, sp^2)

5. 

A & B } ⇒ diastereomers
A & C }
B & C ⇒ enantiomers

Mathematics

PART – III

SECTION – A

1. Six letters can be arranged in following ways: 1, 1, 1, 3 or 1, 1, 2, 2
(1, 1, 1, 3) selection of boxes can be done as ${}^3C_1 \times 2$
(1, 1, 2, 2) selection of boxes can be done as ${}^3C_1 \times {}^3C_2$
$\therefore$ Total such selections = 6 + 9 = 15
$\therefore$ The number of required arrangements = $15 \times 6!$ = 10,800

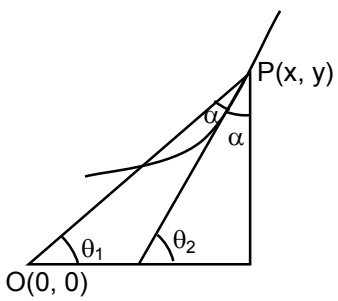
2. Distance between two roots = $\sqrt{b^2 - 4ac}$
Required $-1 \le b^2 - 4ac \le 1$
That means that the point P(b, c) lies on the intersection of the region in between two parabola $x^2 = 4y - 1$ and $x^2 = 4y + 1$ and the square formed by x = 0, x = 1, y = 0 and y = 1
Required probability = $\dfrac{\int_0^1 \dfrac{x^2+1}{4}dx}{1} = \dfrac{1}{3}$

3. $\begin{cases}(2x + y - 1) + \lambda(x - 2y + 3z) = 0 \\ (3x - y + z + 2) + \mu(4x + 5y - 2z - 3) = 0\end{cases}$
Line is parallel to $\dfrac{x}{1} = \dfrac{y}{2} = \dfrac{z}{3}$
So, $(2 + \lambda) \times 1 + (1 - 2\lambda) \times 2 + 3\lambda \times 3 = 0$
$\lambda = -\dfrac{2}{3}$
$(3 + 4\mu) \times 1 + (-1 + 5\mu) \times 2 + (1 - 2\mu) \times 3 = 0$
$\mu = -\dfrac{1}{2}$

4. p, q, r will be in H.P.
$\Rightarrow p + r = pr$
$\Rightarrow \dfrac{pr}{2} \ge \sqrt{pr}$
$\Rightarrow pr \ge 4$

5. Put $x = \dfrac{1}{t}$

6. $\theta_1 + \alpha = \theta_2$
$\theta_1 + 2\alpha = \dfrac{\pi}{2}$
$\theta_1 + 2(\theta_2 - \theta_1) = \dfrac{\pi}{2}$
$2\theta_2 - \theta_1 = \dfrac{\pi}{2}$
$\dfrac{2\tan(\theta_2)}{1 - (\tan(\theta_2))^2} = -\dfrac{1}{\tan\theta_1}$



$$\frac{2\frac{dy}{dx}}{1-\left(\frac{dy}{dx}\right)^2} = -\frac{x}{y}$$

$$2y\frac{dy}{dx} = +x\left(\frac{dy}{dx}\right)^2 - x \Rightarrow x\left(\frac{dy}{dx}\right)^2 - 2y\frac{dy}{dx} = x$$

7. $f(-2) = 3$ and $f(-3)$ is negative
So, equation has one real root in between $(-3, -2)$

8. $f(x) = \frac{2^x+1}{4^x-1} = \frac{1}{2^x-1}$ is a one-one function.

10. Either $\tan x + \tan 2x + \tan 3x = 0$ or $\tan x = \tan 2x = \tan 3x$

11. Major axis will be $x + y = 0$ and centre at (0, 0). Foci will be at distance $\sqrt{3}$ from centre on major axis

12. Circle is $x^2 + y^2 + 8x + 8y + 16 + k(x + y + 12) = 0$
From origin chord of contact is $T = 0$
$\Rightarrow (8 + k_1)x + (8 + k_1)y + 8(4 + 3k_1) = 0$
Which is same as $x + y + 12 = 0$
$\Rightarrow \frac{8+k_1}{1} = \frac{8(4+3k_1)}{12} \Rightarrow k_1 = \frac{16}{3}$
As PQ is chord to $x^2 + y^2 + 8x + 8y + 16 + \frac{16}{3}(x + y + 12) = 0$
From mid-point of PQ(h, k) chord is $T = S_1 \Rightarrow (h, k) = (-6, -6)$

13.-14. The equation of line L is $y = 4x - 1$ and the value of $p = 1$, $q = 6$ and $r = 0$

15.-16. $|A| = 0 \Rightarrow$ A is singular
$|B| = 0 \Rightarrow$ B is singular
$|C| = -1 \Rightarrow$ C is non-singular

$$A^2 = \begin{vmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{vmatrix} = A \Rightarrow A \text{ is idempotent}$$

$$B^2 = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} \Rightarrow B \text{ is nilpotent}$$

$$C^2 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = I \Rightarrow C \text{ is involutary}$$

Now, check the options

17. Equation of the given circle can be written as
$(x^2 + y^2 - 2y - 8) - 2ax = 0$
which represent the family of circle passing through points of intersection of $x^2 + y^2 - 2y - 8 = 0$ and $2ax = 0$.
Now, $2ax = 0$; $x = 0$ put in circle we get
$y^2 - 2y - 8 = 0 \Rightarrow (y - 1)^2 = 9$
$y = 1 \pm 3$
$y = -2, 4$
points can $(0, -2)$, $(0, 4)$

18. Let the tangent at P and Q to a member of this family intersect at (h, k), then PQ is the chord of contact of (h, k) and its equation is
$hx + ky - a(x + h) - (y + k) - 8 = 0$
$x(h - a) + y(k - 1) - (ah + k + 8) = 0$
Comparing this with equation $x = 0$ of PQ. We get
$k = 1$ and $ah + k + 8 = 0$
Since (h, k) lies on the given line $2x + y + 5 = 0$
$2h + 1 + 5 = 0$
$h = 3$
$-3a + 1 + 8 = 0$
$a = 3$
Hence the equation of the required member c of this family is $x^2 + y^2 - 6x - 2y - 8 = 0$

SECTION – C

1. Let the number of rows in triangle is k
$\Rightarrow 1 + 2 + 3 ..... k = n$
$\Rightarrow n = \dfrac{k(k+1)}{2}$
Also, $\dfrac{k(k+1)}{2} + 49 = (k+3)^2$
$\Rightarrow k^2 + k + 98 = 2k^2 + 12k + 18$
$\Rightarrow k^2 + 11k - 80 = 0$
$k^2 + 16k - 5k - 80 = 0$
$\Rightarrow k = 5$
$\Rightarrow$ Number of balls, $n = \dfrac{5(5+1)}{2} = 15$

3. $P = \left|-\left(\dfrac{\cos x}{x+4}\right)\right|_0^{\pi} - \int_0^{\pi} \dfrac{\sin x}{x+4} dx$, put $x = 2t$
$P = \dfrac{1}{\pi+4} + \dfrac{1}{4} - \int_0^{\pi/2} \dfrac{\sin 2t}{t+2} dt$

4. $T_r = \dfrac{\sin r^o \sin(r+1)^o}{\cos r^o \cos(1+r)^o} + 1 - 1 = \dfrac{\cos 1^o}{\cos r^o \cos(r+1)^o} - 1 = \dfrac{\cos 1^o}{\sin 1^o}\left\{\dfrac{\sin\{(r+1)^o - r^\circ\}}{\cos r^o \cos(r+1)^o}\right\} - 1$
$= \cot 1^o \{\tan(r+1)^o - \tan r^o\} - 1 \Rightarrow T_1 + T_2 + ...T_{88} = \cot 1^o(\tan 89^o - \tan 1^o) - 88 = \cot^2(1^o) - 89$

5. $X + iy = \dfrac{a_r + 2i}{(a_r)^2 + 4}$

$X = \dfrac{a_r}{a_r^2 + 4}$, $Y = \dfrac{2}{a_r^2 + 4}$

$X^2 + Y^2 = \dfrac{1}{2}Y$

$X^2 + Y^2 - \dfrac{1}{2}Y = 0$

So all points lies on circle $x^2 + y^2 - \dfrac{y}{2} = 0$

$r = \dfrac{1}{4}$

Area of regular octagon = $8\left(\dfrac{1}{2} \times r^2 \times \sin\left(\dfrac{2\pi}{8}\right)\right) = 8 \times \dfrac{1}{2} \times \dfrac{1}{16} \times \dfrac{1}{\sqrt{2}} = \dfrac{1}{4\sqrt{2}}$